

Are electrophysiological and oligodendrocyte alterations an element in the development of multiple sclerosis at the same time as or before the immune response?

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Efficient communication between the glial cells and neurons is a bi-directional process that is essential for conserving normal functioning in the central nervous system (CNS). Neurons dynamically regulate other brain cells in the healthy brain, yet little is known about the first pathways involving oligodendrocytes and neurons. Oligodendrocytes are the myelin-forming cells in the CNS that are needed for the propagation of action potentials along axons and additionally serve to support neurons by neurotrophic factors (NFTs). In demyelinating diseases, like multiple sclerosis (MS), oligodendrocytes are thought to be the victims. Axonal damage begins early and remains silent for years, and neurological disability develops when a threshold of axonal loss is reached, and the compensatory mechanisms are depleted. Three hypotheses have been proposed to explain axonal damage: 1) the damage is caused by an inflammatory process; 2) there is an excessive accumulation of intra-axonal calcium levels; and, 3) demyelinated axons evolve to a degenerative process resulting from the lack of trophic support provided by myelin or myelin-forming cells. Although MS was traditionally considered to be a white matter disease, the demyelination process also occurs in the cerebral cortex. Recent data supports the notion that initial response is triggered by CNS injury. Thus, the understanding of the role of neuron-glial neurophysiology would help provide us with further explanations. We should take in account the suggestion that MS is in part an autoimmune disease that involves genetic and environmental factors, and the pathological response leads to demyelination, axonal loss and inflammatory infiltrates.

Analysis of the spatial and morphological characteristics of oligodendrocytes from images of in vitro culture.

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Oligodendrocytes (OLs) are glial cells responsible for the formation of myelin sheaths in the central nervous system. The characteristic features of the oligodendrocyte lineage, ranging from proliferative and migratory oligodendrocyte progenitor cells (OPC) to myelinating mature OLs, can be observed in vitro cultures of OL lineage cells. Here, we introduce a method for analyzing the spatial distribution of OPCs, which reflects their capacity for proliferation and migration, and the morphological complexity of mature OLs, which reflects their capacity for myelin formation, from immunostaining images of in vitro OL cultures. Through the methods described, we have demonstrated the tendency for OPCs to cluster in an environment with epidermal growth factor (EGF), and the differing morphological complexity of mature OLs according to culture medium and duration of differentiation. •The proliferative and migratory characteristics of OPCs can be evaluated by analyzing their spatial distribution. •The myelin-forming capacity of mature OLs can be measured by analyzing their morphological complexity. •Image-based analyses may be a substitute for more convoluted experiments to assess OL function.

Effect of central myelin on the proliferation and differentiation into O4(+) oligodendrocytes of GFP-NSCs.

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Myelin is a biologically active multilamellar that is formed by oligodendrocytes (OLs) in the central nervous system (CNS) and ensheathes axons (Ishii et al., Proc Natl Acad Sci USA 106:14605-14610, 2009). Myelin damage is related to neurological trauma such as spinal cord injury (SCI). In this article, we investigated whether myelin derived from rat spinal cord can influence the proliferation of neural stem cells (NSCs) and NSCs differentiation into oligodendrocytes in vitro. After extracting myelin, we verified that myelin preparation was successful by western blot analysis. Then, we explored the effects of different myelin concentrations on the NSCs proliferation by MTT assays. Our results showed that 2 µg/ml myelin can promote the proliferation of NSCs, while NgR antibody can antagonize the effect. In addition, myelin can inhibit the differentiation of NSCs into O4(+) oligodendrocytes impeding them maturation. In conclusion, these results suggested that central myelin can affect the proliferation and differentiation of NSCs, thus promoting us to understand further the complex roles of myelin in NSCs after CNS injury.

Tissue transglutaminase activity is involved in the differentiation of oligodendrocyte precursor cells into myelin-forming oligodendrocytes during CNS remyelination.

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During normal brain development, axons are myelinated by mature oligodendrocytes (OLGs). Under pathological, demyelinating conditions within the central nervous system (CNS), axonal remyelination is only partially successful because oligodendrocyte precursor cells (OPCs) largely remain in an undifferentiated state resulting in a failure to generate myelinating OLGs. Tissue Transglutaminase (TG2) is a multifunctional enzyme, which amongst other functions, is involved in cell differentiation. Therefore, we hypothesized that TG2 contributes to differentiation of OPCs into OLGs and thereby stimulates remyelination. In vivo studies, using the cuprizone model for de- and remyelination in TG2(-/-) and wild-type mice, showed that during remyelination expression of proteolipid protein mRNA, as a marker for remyelination, in the corpus callosum lags behind in TG2(-/-) mice resulting in less myelin formation and, moreover, impaired recovery of motor behavior. Subsequent in vitro studies showed that rat OPCs express TG2 protein and activity which reduces when the cells have matured into OLGs. Furthermore, when TG2 activity is pharmacologically inhibited, the differentiation of OPCs into myelin-forming OLGs is dramatically reduced. We conclude that TG2 plays a prominent role in remyelination of the CNS, probably through stimulating OPC differentiation into myelin-forming OLGs. Therefore, manipulating TG2 activity may represent an interesting new target for remyelination in demyelinating diseases.

[Development and regeneration of oligodendrocytes: therapeutic perspectives in demyelinating diseases].

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The function of the central nervous system (CNS) is in great part depending on glial cells as, for instance, radial glial cells give rise to cortical neurons, and oligodendrocytes synthesize an immense specialized membrane that enwraps axons to make myelin internodes. Myelin allows fast saltatory conduction of action potentials along myelinated nerve tracts and assures the survival of axons. Oligodendrocytes precursors (OP) emerge during development, first in the spinal cord and later in the telencephalon from multipotential neural precursors in germinative zones around the cerebral ventricles. Morphogens and specific growth factors stimulate the growth, migration and survival of OPs toward axons, culminating in

myelination. Such precursors can be isolated from human brain and persist in the adult CNS, allowing some degree of remyelination in the course of a demyelinating disease caused by an infectious agent or inflammation such as multiple sclerosis (MS). These remyelinating cells can recapitulate some molecular events of myelination while new OPs are generated by neural stem cells in the subventricular zones and niches. This natural repair process often decreases with time in man, raising questions about the appropriateness of rodent animal models where remyelination is robust. The challenge today in MS is to develop a pharmacology of myelin repair by endogenous precursors which, if successful, might be more likely to result in clinical benefits than transplantation of myelin-forming cells, shown to be so efficient in rodent models.

FBXW7 regulates MYRF levels to control myelin capacity and homeostasis in the adult CNS.

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Myelin, along with the oligodendrocytes (OLs) that produce it, is essential for proper central nervous system (CNS) function in vertebrates. Although the accurate targeting of myelin to axons and its maintenance are critical for CNS performance, the molecular pathways that regulate these processes remain poorly understood. Through a combination of zebrafish genetics, mouse models, and primary OL cultures, we found FBXW7, a recognition subunit of an E3 ubiquitin ligase complex, is a regulator of adult myelination in the CNS. Loss of Fbxw7 in myelinating OLs resulted in increased myelin sheath lengths with no change in myelin thickness. As the animals aged, they developed progressive abnormalities including myelin outfold, disrupted paranodal organization, and ectopic ensheathment of neuronal cell bodies with myelin. Through biochemical studies we found that FBXW7 directly binds and degrades the N-terminal of Myelin Regulatory Factor (N-MYRF), to control the balance between oligodendrocyte myelin growth and homeostasis.

RhoA regulates oligodendrocyte differentiation and myelination by orchestrating cortical and membrane tension.

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Timely differentiation and myelin formation by oligodendrocytes are essential for the physiological functioning of the central nervous system (CNS). While the Rho GTPase RhoA has been hinted as a negative regulator of myelin sheath formation, the precise in vivo mechanisms have remained elusive. Here we show that RhoA controls the timing and progression of myelination by oligodendrocytes through a fine-tuned balance between cortical tension, membrane tension and cell shape. Using a conditional mouse model, we observe that Rhoa ablation results in the acceleration of myelination driven by hastened differentiation and facilitated through membrane expansion induced by changes in MLCII activity and in F-actin redistribution and turnover within the cell. These findings reveal RhoA as a central molecular integrator of alterations in actin cytoskeleton, actomyosin contractility and membrane tension underlying precise morphogenesis of oligodendrocytes and normal myelination of the CNS.

Altered Translational Control of Fragile X Mental Retardation Protein on Myelin Proteins in Neuropsychiatric Disorders.

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Myelin is a specialized structure of the nervous system that both enhances electrical conductance and insulates neurons from external risk factors. In the central nervous system, polarized oligodendrocytes form myelin by wrapping processes in a spiral pattern around neuronal axons through myelin-related gene regulation. Since these events occur at a distance from the cell body, post-transcriptional control of gene expression has strategic advantage to fine-tune the overall regulation of protein contents in situ. Therefore, many research interests have been focused to identify RNA binding proteins and their regulatory mechanism in myelinating compartments. Fragile X mental retardation protein (FMRP) is one such RNA binding protein, regulating its target expression by translational control. Although the majority of works on FMRP have been performed in neurons, it is also found in the developing or mature glial cells including oligodendrocytes, where its function is not well understood. Here, we will review evidences suggesting abnormal translational regulation of myelin proteins with accompanying white matter problem and neurological deficits in fragile X syndrome, which can have wider mechanistic and pathological implication in many other neurological and psychiatric disorders.

Behaviour of oligodendrocytes and Schwann cells in an experimental model of toxic demyelination of the central nervous system.

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Oligodendrocytes and Schwann cells are engaged in myelin production, maintenance and repairing respectively in the central nervous system (CNS) and the peripheral nervous system (PNS). Whereas oligodendrocytes act only within the CNS, Schwann cells are able to invade the CNS in order to make new myelin sheaths around demyelinated axons. Both cells have some limitations in their activities, i.e. oligodendrocytes are post-mitotic cells and Schwann cells only get into the CNS in the absence of astrocytes. Ethidium bromide (EB) is a gliotoxic chemical that when injected locally within the CNS, induce demyelination. In the EB model of demyelination, glial cells are destroyed early after intoxication and Schwann cells are free to approach the naked central axons. In normal Wistar rats, regeneration of lost myelin sheaths can be achieved as early as thirteen days after intoxication; in Wistar rats immunosuppressed with cyclophosphamide the process is delayed and in rats administered cyclosporine it may be accelerated. Aiming the enlightening of those complex processes, all events concerning the myelinating cells in an experimental model are herein presented and discussed.

Chromatin remodeling in oligodendrogenesis.

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Oligodendrocytes are one type of glial cells responsible for myelination and providing trophic support for axons in the central nervous system of vertebrates. Thanks to myelin, the speed of electrical-signal conduction increases several hundred-fold because myelin serves as a kind of electrical insulator of nerve fibers and allows for quick saltatory conduction of action potentials through Ranvier nodes, which are devoid of myelin. Given that different parts of the central nervous system are myelinated at different

stages of development and most regions contain both myelinated and unmyelinated axons, it is obvious that very precise mechanisms must exist to control the myelination of individual axons. As they go through the stages of specification and differentiation - from multipotent neuronal cells in the ventricular zone of the neural tube to mature myelinating oligodendrocytes as well as during migration along blood vessels to their destination - cells undergo dramatic changes in the pattern of gene expression. These changes require precisely spatially and temporally coordinated interactions of various transcription factors and epigenetic events that determine the regulatory landscape of chromatin. Chromatin remodeling substantially affects transcriptional activity of genes. The main component of chromatin is the nucleosome, which, in addition to the structural function, performs a regulatory one and serves as a general repressor of genes. Changes in the type, position, and local density of nucleosomes require the action of specialized ATP-dependent chromatin-remodeling complexes, which use the energy of ATP hydrolysis for their activity. Mutations in the genes encoding proteins of the remodeling complexes are often accompanied by serious disorders at early stages of embryogenesis and are frequently identified in various cancers. According to the domain arrangement of the ATP-hydrolyzing subunit, most of the identified ATP-dependent chromatin-remodeling complexes are classified into four subfamilies: SWI/SNF, CHD, INO80/SWR, and ISWI. In this review, we discuss the roles of these subunits of the different subfamilies at different stages of oligodendrogenesis.
